

6.2 Equivalent Photon Approximation

Source: Peskin & Schroeder, *An Introduction to Quantum Field Theory*, Problem 6.2.

Consider the process in which electrons of very high energy scatter from a target. In leading order in α , the electron is connected to the target by one photon propagator:

$$i\mathcal{M} = \quad (1.1)$$

where the blob is, in general, complicated coupling of the virtual photon to the target.

If the initial and final energies of the electron are E and E' . Under the high-energy limit,

$$p = (E, 0, 0, E), \quad p' = (E', E' \sin \theta, 0, E' \cos \theta), \quad (1.2)$$

which implies

$$q^2 = (p' - p)^2 = -2EE'(1 - \cos \theta). \quad (1.3)$$

In the limit of forward scattering, whatever the energy loss, the photon momentum approaches $q^2 = 0$; thus the reaction is highly peaked in the forward direction. It is tempting to guess that, in this limit, the virtual photon becomes a real photon. Let us investigate in what sense that is true.

(a) The matrix element of the above process is given by

$$\mathcal{M} = (-ie)\bar{u}(p')\gamma^\mu u(p)\frac{-ig_{\mu\nu}}{q^2}\hat{\mathcal{M}}^\nu(q), \quad (1.4)$$

where $\hat{\mathcal{M}}^\nu(q)$ represents the (in general, complicated) coupling of the virtual photon to the target. Let us analyse the structure of the piece $\bar{u}(p')\gamma^\mu u(p)$. Let $q = (q^0, \vec{q})$ and $\tilde{q} = (q^0, -\vec{q})$. We can expand the spinor products as

$$\bar{u}(p')\gamma^\mu u(p) = Aq^\mu + B\tilde{q}^\mu + C\epsilon_1^\mu + D\epsilon_2^\mu, \quad (1.5)$$

where A, B, C, D are functions of the scattering angle and energy loss and ϵ_1, ϵ_2 are two unit vectors transverse to $\vec{q}$. By dotting the expression with q^μ , we find

$$q_\mu \cdot [\bar{u}(p')\gamma^\mu u(p)] = Aq^2 + Bq \cdot \tilde{q}. \quad (1.6)$$

By Ward identity, we have $q_\mu \bar{u}(p') \gamma^\mu u(p) = 0$. Thus, we have

$$B = -A \frac{q^2}{q \cdot \tilde{q}} \approx A \frac{2EE'(1 - \cos \theta)}{(q^0)^2 + \vec{q}^2} \sim \theta^2. \quad (1.7)$$

Thus, B can be ignored in the case of forward scattering.

Again, by Ward identity, we have $q \cdot \hat{\mathcal{M}}(q) = 0$. Thus, A can be ignored as well.

(b) In the frame where $p = (E, 0, 0, E)$, the spinor $u_s(p)$ is given by

$$u_s(p) = \begin{pmatrix} \sqrt{p \cdot \vec{\sigma}} \xi_s \\ \sqrt{p \cdot \vec{\sigma}} \xi_s \end{pmatrix} \quad (1.8)$$

where

$$\sigma^3 \xi_\pm = \pm \xi_\pm \implies \xi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \xi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (1.9)$$

and

$$\sqrt{p \cdot \vec{\sigma}} = \sqrt{E(1 - \sigma^3)}, \quad \sqrt{p \cdot \vec{\sigma}} = \sqrt{E(1 + \sigma^3)}. \quad (1.10)$$

Thus,

$$\boxed{u_+(p) = \sqrt{2E} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_-(p) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.} \quad (1.11)$$

Next, let us evaluate $u(p')$. Using $p' = (E', E' \sin \theta, 0, E' \cos \theta)$, we find

$$\hat{p}' \cdot \vec{\sigma} \xi_\pm = \pm \xi_\pm \implies \xi_+ = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix}, \quad \xi_- = \sqrt{2E'} \begin{pmatrix} -\sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} \end{pmatrix} \quad (1.12)$$

and

$$\sqrt{p' \cdot \vec{\sigma}} = \sqrt{E'(1 - \hat{p}' \cdot \vec{\sigma})}, \quad \sqrt{p' \cdot \vec{\sigma}} = \sqrt{E'(1 + \hat{p}' \cdot \vec{\sigma})}. \quad (1.13)$$

Thus, we find

$$\boxed{u_+(p') = \sqrt{2E'} \begin{pmatrix} 0 \\ 0 \\ \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix}, \quad u_-(p') = \sqrt{2E'} \begin{pmatrix} -\sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} \\ 0 \\ 0 \end{pmatrix}.} \quad (1.14)$$

The polarisation vectors are defined to be parallel and horizontal to the scattering plane ($x - z$ plane). Thus,

$$\epsilon_1^\mu = \frac{1}{\sqrt{E^2 + E'^2 - 2EE' \cos \theta}} (0, E' \cos \theta - E, 0, -E' \sin \theta), \quad \epsilon_2^\mu = (0, 0, 1, 0). \quad (1.15)$$

Now, let us evaluate $\bar{u}(p') \vec{\gamma} \cdot \vec{\epsilon}_i u(p)$. With some algebra, one should find

$$\begin{aligned} \bar{u}_\pm(p') \vec{\gamma} \cdot \vec{\epsilon}_1 u_\pm(p) &= -\frac{E' + E}{|E' - E|} \theta \sqrt{EE'} = C, \\ \bar{u}_\pm(p') \vec{\gamma} \cdot \vec{\epsilon}_2 u_\pm(p) &= \pm \theta \sqrt{EE'} = D_\pm, \\ \bar{u}_\pm(p') \vec{\gamma} \cdot \vec{\epsilon}_1 u_\mp(p) &= \bar{u}_\pm(p') \vec{\gamma} \cdot \vec{\epsilon}_2 u_\mp(p) = 0. \end{aligned} \quad (1.16)$$